

How the T20 Power Rankings Are Calculated

A power ranking answers a different question from the league table. The table asks **"who has earned the most points?"** The power ranking asks **"who is playing the best cricket right now?"** — so a team can sit fourth in the table and still top this list.

The founding rule: the maths decides the order, the AI does not. Every position is computed by fixed formulas from match data. The language model is only allowed to write the sentence explaining a team's position — it never sees, chooses, or influences the rank. Every ranking is reproducible, and every place is answerable with a number.

Part 1 — In plain English

Think of it as a **school report card**. Instead of one exam, each team is graded on nine subjects. Every subject is marked out of 100, some subjects count more than others, and the marks are combined into one final score. Sort by that score, and you have the rankings.

#	WHAT WE MEASURE	WHAT IT TELLS YOU	COUNTS FOR
1	Win %	How often they win. The foundation.	30%
2	Death overs (16–20)	The closing overs, where T20 games are decided: their hitting versus their bowling.	15%
3	Rolling Net Run Rate	Whether they out-score opponents, over the last 5 games only , so one early thrashing doesn't flatter them all season.	15%
4	Powerplay (overs 1–6)	Command of the opening burst — runs scored there minus runs conceded.	12%
5	Momentum (form)	Are they hot <i>right now</i> ? The latest match counts most.	12%
6	Home / away	Teams that only win at home are flattered; travelling well counts.	5%
7	Key players	A side missing its stars is not the same side, whatever last week's results say.	5%
8	Win quality	<i>How</i> they win and lose — a 60-run rout versus a last-ball escape.	4%
9	Strength of schedule	Whether the opponents faced were any good.	2%

Where the weights come from. They are set by cricket judgement, not by a training run. Win% leads because winning is the point. Net run rate and the two phase measures — the powerplay and the death overs — come next, because they capture *how* a side controls a game rather than just the result. Form, home/away, key players and win quality fill in the margins. They are deliberately round numbers, and they are published here so anyone can argue with them.

Why not fit them statistically? We tried. Weights tuned against simulated seasons scored better on the simulator, but the simulator built each team's powerplay and death-overs numbers straight out of a single hidden "strength" value — so those two measures were close to reading the answer key, and the fit handed them a third of the table. Real teams have phase *styles* that do not collapse to one number. Rather than publish a precision we cannot support, the weights are honest editorial choices, and we will re-fit only when a full season of real results exists to fit against.

Why momentum isn't simply "the last five". Recent matches count far more than old ones, fading smoothly rather than stopping dead at a cut-off. In practice the most recent match carries about 35% of this grade, and the last five together about 88%.

Where a season starts. On the morning of the first match there are no results to grade, so every subject sits at its neutral mark and all seven teams score an identical 34.0 — which would say the reigning champions and the bottom side are the same team. They are not. Each team therefore **opens on the rating it earned last season**, computed by running this same method over the completed 2025 season, and that inherited rating is then worth less with every match actually played this year: half of it is gone after four games, roughly three-quarters after twelve, and by the back end of a season the table is entirely 2026 form. Jamaica Kingsmen are new and have no 2025 record to inherit, so they open one standard deviation below the 2025 league average — a first-year discount that is stated rather than hidden. Press **Fresh Start** to switch the carry-in off and rank on this season alone.

How much last season is trusted. A record describes the squad that produced it, and these squads changed a great deal over the winter — Saint Lucia kept barely a third of theirs, Barbados two thirds. A team that has largely been rebuilt is therefore held closer to the league average than one that kept its side together: last season still counts, but it counts less when fewer of the players are still there. Note what this is and is not. It measures *how much we know*, not *how good the new squad is* — so a weak side that rebuilt also moves toward the middle, because its poor record has stopped being reliable evidence too.

The score and the arrows. The nine graded subjects combine into one index, which is then published on a **70–89 rating scale** — the range broadcast and print audiences are used to seeing. The conversion is a fixed piece of arithmetic applied identically to every team, so it changes the numbers on the page but never the order. The **▲/▼ arrows** show movement against the previous published ranking: **▲2** means two places gained, and a team that has not moved simply shows nothing.

Nothing here is opinion. Most published power rankings are a writer's judgement call. This one is arithmetic: run the same matches through it twice and you get the same table, every time.

Part 2 — The technical method

Pipeline. `data` → `collect()` → nine metrics → normalise to `[0,1]` → `weighted sum × 100` → `sort`. Weights and bounds live in `weights.config.json`, never in code. Implementation: `src/engine.js`.

Notation. For team *t*: *W/P* = wins/played; *RF*, *OF* = runs scored and overs faced; *RA*, *OA* = runs and overs conceded. Every metric derives from raw per-innings data — nothing is hand-entered.

The nine components

1. Win percentage — *W / P*

2. Death-overs net (*overs 16–20*)

```
net = deathBatSR - deathBowlEcon × 16.67
```

The factor `100/6 = 16.67` converts bowling economy (runs/over) onto the batting strike-rate scale (runs per 100 balls), making the two dimensionally comparable.

3. Rolling NRR (*last 5 matches*)

```
rNRR = (Σ RF / Σ OF) - (Σ RA / Σ OA)
```

Per ICC convention a side **bowled out** is charged its full 20-over quota rather than the overs it actually faced, so being dismissed cheaply cannot flatter a run rate. A side that wins a chase early keeps the overs it used. Also reported: `trend = rolling - season`.

4. Powerplay dominance (*overs 1–6, i.e. 6 overs per innings per match*)

```
PP = (ppRunsFor / 6P) - (ppRunsAgainst / 6P) [runs per over]
```

5. Momentum — exponentially weighted update over **every** match played, seeded at 0.5:

```
m_t = α · (win ? 1 : 0) + (1 - α) · m_{t-1} , α = 0.35
```

Match weights decay 0.350, 0.228, 0.148, 0.096, 0.063... — a **half-life of ≈1.6 matches**.

6. Home/away — `0.6 · awayWin% + 0.4 · homeWin%`, deliberately over-weighting away form.

7. Key-player availability — `starsAvailable / squadStars`, taken **as of the most recent match**.

8. Win quality — mean **signed runs margin over every match**, wins and losses alike:

```
defended : own - opp (runs)
chased   : (120 - balls used) × run rate (runs equivalent)
lost     : the same quantity, negated
```

Both outcomes are expressed in one unit — runs — so defending and chasing are directly comparable, and heavy defeats cost what heavy wins earn. Normalised over ±40 runs.

9. Strength of schedule — mean opponent win rate, **excluding their matches against this team**. Counting head-to-head made it a mirror of the team's own record (every win you take is a loss on an opponent's card), which inverted the metric. In a complete round robin every side then faces the identical field, so the value is the same for everyone and changes no ranking; it only carries information on an uneven schedule.

Normalisation and the final score

Each raw metric maps onto `[0,1]` via `norm(x) = clamp((x - min)/(max - min))`. Ratios (1, 6, 7, 9) are already 0–1; the rest use configured bounds — **death net ±60, rNRR ±2.00, powerplay ±3.00 rpo, win quality ±40 runs**. Then:

```
score = 100 × Σ ( w_k · n_k )
```

weights `0.30, 0.15, 0.15, 0.12, 0.12, 0.05, 0.05, 0.04, 0.02`, summing to exactly 1.00. **Ties** break on win% then rolling NRR.

Delta = previous rank – current rank.

Season carry-in (the prior)

With no matches played, every normalised metric falls to its neutral value and the weighted sum is identically `34.0` for every team. Ranking a new season from that point discards the only evidence that exists. So each team opens on the index it earned in the previous season and that opening claim decays as real results arrive:

```
prior_weight = k / (k + played) k = prior.halfLifeMatches (default 4)
score        = prior_weight · prior + (1 - prior_weight) · thisSeasonIndex
```

At `played = 0` the weight is exactly 1, so the published rating **is** last season's index; at `played = k` the prior carries half; at ten matches about 29%. The unblended in-season index stays on every row (`inSeasonScore`) so an audit can always see what this season alone says.

Squad continuity. A record only describes the squad that produced it. Between 2025 and 2026, 23 players switched franchise and 33 left the league, and it fell unevenly — Saint Lucia kept 39% of their squad, Barbados 67%. So each prior is pulled toward the league mean in proportion to how much of the side is gone:

```
λ      = confidenceFloor + (1 - confidenceFloor) · kept / squad2025
prior' = leagueMean + λ · (prior - leagueMean)
```

with `confidenceFloor = 0.5`, so even a fully rebuilt squad keeps half its distance from the mean — coaching, conditions and a core identity carry over. Per-team counts live in `data/cpl_2026.json → squadMovement`, taken from the season review's movement tables, and reconcile exactly with its published totals.

This is **uncertainty, not punishment**, and the direction follows from that: heavy churn makes a *bad* record less predictive too, so a below-average side that rebuilt moves *up* toward the mean. Saint Lucia — most churned of the seven — gains slightly despite losing their captain and their leading spinner. Making churn actively cost a team would mean valuing the individual players who left, which is a judgement call, and the ordering is meant to be arithmetic. Teams with no previous season are skipped: Jamaica's standing is already set by the expansion rule.

The prior is **derived, not entered**: `src/priors.js` runs this same engine with these same weights over `data/cpl_2025.json`, so re-tuning a weight moves the carry-in and the live table together. A rebranded franchise inherits its predecessor's index via `priorFrom` in the season data — Barbados Tridents carry Barbados Royals' 2025 season. A genuinely new franchise cannot be measured, so it opens `prior.expansionSigma` standard deviations below the previous league mean; at the default 1.0 that puts Jamaica Kingsmen sixth of seven.

Two honest limits. The 2025 source carries no ball-by-ball data, so the powerplay and death-overs terms evaluate to exactly 0.5 for all six teams in the 2025 pass — they contribute an identical 13.5 points to every prior and therefore shift the scale but **cannot change the 2025 order**. And the carry-in openly *does* change the published order; that is its purpose. Fresh Start turns it off.

Published rating. The raw index is what every calculation and audit record uses; graphics carry a presentation figure mapped from it — `rating = 70 + (index-15)/70 × 19`, clamped to `[70, 89]`. The constants are fixed in config, not derived from the current table, so ratings are comparable week to week, and the map is strictly increasing — a change of units, never of order.

Choosing the weights. They are editorial, set by cricket judgement and stored in `weights.config.json` so they can be changed without touching the engine. An earlier set was fitted by maximising rank correlation against latent team strength over 300 synthetic seasons; that fit was withdrawn because the generator derived phase splits directly from strength, so it over-credited the powerplay and death-overs terms. The fitting method transfers unchanged to real data — only the target needs to become actual results.

Optional metrics (off by default)

Five extras can be enabled per session: **expected wins** (Pythagorean $RF^2 / (RF^2 + RA^2)$), **toss independence** (win% when *losing* the toss), **chase/set versatility**, **top-4 consistency**, and **bowling spread** (inverse share of wickets taken by the top two bowlers). Enabling them pushes total weight above 1.0, so scores scale up while the ordering stays valid.

Known limitations — stated deliberately

- **The weights are judgement, not evidence.** They are reasonable and openly published, but no live data yet proves this particular split is optimal. Re-fit against a full season of real results when one exists.
- **Correlated components.** Several measures move together — home/away tracks Win% very closely, and form overlaps rolling NRR — so the nine subjects are less independent than the table implies, and Win% effectively carries more than its stated 30%.
- **Key players** is a supplied figure, not yet a live ICC top-30 feed, so it reflects whatever availability was entered. It is also a *ratio* of available to squad stars, which means it is **1.0 for everybody at the start of a season** — the `squadStars` numbers change nothing on an opening table, and the term only does work once a match reports someone missing. Squad *strength* reaches the table through continuity instead.
- **Squad continuity counts heads, not quality.** Losing a leading wicket-taker weighs exactly the same as losing a fringe seamer. Weighting departures by what those players contributed would need per-player valuations, which is the judgement call the ordering is designed to exclude.
- **T20-shaped.** A 120-ball innings and a 6-over powerplay are assumed. Rain-shortened matches and other formats are not modelled, and the powerplay term assumes six powerplay overs per innings.
- **Strength of schedule is inert in a balanced league.** By design it is identical for every team in a complete round robin; it only does work on an uneven schedule.
- **The season carry-in is an assumption, not a measurement.** Last season is evidence about this one, but it is not proof: squads change, and the 2025 phase terms are constants because no ball-by-ball source was available. The decay rate (`prior.halfLifeMatches = 4`) and the expansion discount (`prior.expansionSigma = 1.0`) are editorial settings, published here so they can be argued with, and both are switched off entirely by Fresh Start.

Computed by `src/engine.js`; weights tunable in `weights.config.json` without code changes. Full fitting write-up in `parameters.md`. Every generated blurb is fact-checked against these numbers and logged with its sources for editorial review.